



For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory only.

CIPRODAC 500 TABLET

Composition:

Each film-coated tablet contains:
Ciprofloxacin Hydrochloride USP
equivalent to Ciprofloxacin 500 mg
White-coloured, special capsule-shaped, smooth film-coated tablet, plain on both sides.

Actions and Pharmacology:

Ciprofloxacin is a synthetic broad spectrum fluoroquinolone antibacterial agent which is active against a wide range of gram-positive and gram negative organisms. Its bactericidal activity is a result of the inhibition of the A-subunit of the enzyme DNAgyrase.

DNA-gyrase is a type II topoisomerase essential for ATP-dependent coiling and supercoiling of bacterial DNA, a process which enables it to be replicated and repacked in both daughter cells. Ciprofloxacin inhibits relaxation of supercoiled DNA and promotes double-stranded DNA breakage.

In vitro studies have shown the susceptibility of most strains of the following organisms:

Gram-negative : Escherichia coli, Klebsiella species (including, respectively, Klebsiella pneumoniae and Klebsiella oxytoca), Enterobacter species, Citrobacter species, Edwardsiella tarda, Salmonella species, Shigella species, Proteus mirabilis, Proteus vulgaris, Providencia stuartii, Providencia rettgeri, Morganella morganii, Serratia species (including Serratia marcescens), Yersinia enterocolitica, Pseudomonas aeruginosa, Acinetobacter species, Haemophilus influenzae, Haemophilus parainfluenzae, Haemophilus ducreyi, Neisseria gonorrhoea, Neisseria meningitidis, Branhamella catarrhalis, Campylobacter species, Aeromonas species, Vibrio species (including V.cholerae), Brucella melitensis, Pasteurella multocida and Legionella species, Brucella species susceptibility is marginal.

Gram-positive: Staphylococcus aureus (including resistant and non-resistant to methicillin), Staphylococcus species (including Staphylococcus epidermidis), Streptococcus pyogenes and Streptococcus pneumoniae. Most Streptococcus strains, including Streptococcus faecalis, are only moderately or marginally susceptible to ciprofloxacin, like Mycobacterium tuberculosis and Chlamydia trachomatis. Ureaplasma urealyticum and Chlamydia trachomatis are at the resistant end of the treatable range with MICs of 1-2 mg/l. Staphylococci are generally moderately or marginally susceptible.

Most Pseudomonas cepacia strains and some Pseudomonas maltophilia strains are resistant to ciprofloxacin, as are anaerobe bacteria including Bacteroides fragilis and Clostridium difficile. Ciprofloxacin is less active when tested in acidic pH. The inoculum size has little effect when tested in vitro. Minimal bactericidal concentration (MBC) is generally not more than double the minimum inhibitory concentration (MIC). Resistance to ciprofloxacin in vitro usually develops slowly (multistep mutation). No enzymes which destroy ciprofloxacin are known. Ciprofloxacin does not generally cross-react with other antimicrobial agents, such as beta-lactams or aminoglycosides, and gentamicin-resistant or β -lactamase-producing bacteria usually remain susceptible to quinolones. In vitro studies have shown that additive activity often results when ciprofloxacin is used in combination with other agents like beta-lactams, aminoglycosides, clindamycin or metronidazole.

Ciprofloxacin is rapidly and well-absorbed after oral administration, and undergoes minimal first-pass metabolism. Presence of food in the gastrointestinal tract decreases the rate but not the extent of absorption of the drug. The oral bioavailability of ciprofloxacin hydrochloride is 50-85% in healthy, fasting adults and peak serum concentrations of the drug generally are attained within 0.5-2.3 hours. Peak serum concentrations and area under the serum concentration time curve (AUC) increase in proportion to the dose over the oral dose range of 250-750mg and are unaffected by gender. Following oral administration of a single 250mg dose of ciprofloxacin in healthy adults, peak serum concentrations average 0.76-1.5 mcg/ml; serum concentrations 12 hours after the dose average 0.1mcg/ml. Peak serum concentration and AUCs are slightly higher in geriatric patients than in younger patients; this may occur because of increased bioavailability, reduced volume of distribution, and/or reduced renal clearance in the patients.

Ciprofloxacin is widely distributed into body tissues and fluids following oral or IV administration. Higher concentrations of the drug generally are attained in bile, lungs, kidney, liver, gallbladder, uterus, seminal fluid, prostatic tissue and fluid, tonsils, endometrium, fallopian tubes and ovaries. Concentrations of the drug achieved in most of these tissues and fluids substantially exceed those in serum.

In healthy adults, the apparent volume of distribution of ciprofloxacin is 2-3.5l/kg and the apparent volume of distribution at steady state is 1.7-2.7l/kg. The apparent volume of distribution of ciprofloxacin in geriatric patients 64-91 years of age averages 3.5-3.6l/kg.

Only low concentrations of ciprofloxacin are distributed into CFS; peak CFS concentrations may be 6-10% of peak serum concentrations.

Ciprofloxacin is 16-43% bound to serum proteins in vitro. Ciprofloxacin crosses the placenta and is distributed into amniotic fluid in humans. The drug is also distributed into milk. In lactating women who received 750mg of ciprofloxacin every 12 hours for 3 doses, milk concentrations were higher than concomitant serum concentrations for up to 12 hours after a dose.

The serum elimination half-life of ciprofloxacin in adults with normal renal functions is 3-5 hours. The elimination half-life of the drug is slightly longer in geriatric adults than in younger adults, and ranges from 3.3-6.8 hours in adults 60-91 years of age with renal function normal for their age.

In patients with impaired renal function, serum concentrations of ciprofloxacin are higher and the half-life prolonged. In adults with creatinine clearances of 30ml/minute or less, half-life of the drug ranges from 4.4-12.6 hours. In contrast, no significant changes in ciprofloxacin pharmacokinetics have been observed in patients with stable chronic liver cirrhosis.

Ciprofloxacin is eliminated by renal and non-renal mechanisms. The drug is partially metabolized in the liver by modification of the piperazinyl group to at least 4 metabolites.

Ciprofloxacin and its metabolites are excreted in urine and faeces. Unchanged ciprofloxacin is excreted in urine by both glomerular filtration and tubular secretion.

Following oral administration of a single 250-, 500-, or 700-mg dose in adults with normal renal function, 15-50% of the dose is excreted in urine as unchanged drug and 10-15% as metabolites within 24 hours; 20-40% of the dose is excreted in faeces as unchanged drug and metabolites within 5 days. Most, but not all, of unchanged ciprofloxacin in faeces appears to result from biliary excretion.

INDICATIONS:

Consideration should be given to available official guidance on the appropriate use of antibacterial agents.

For Ciprofloxacin film-coated tablets and solution for infusion:

Uncomplicated and complicated infections caused by ciprofloxacin susceptible pathogens.

Infections of the respiratory tract:

Ciprofloxacin can be regarded as an advisable treatment for pneumonias caused by Klebsiella, Enterobacter, Proteus, E. coli, Pseudomonas, Haemophilus, Moraxella catarrhalis, Legionella and Staphylococcus.

Infections of the middle ear (otitis media), of the paranasal sinuses (sinusitis), especially if these are caused by Gram-negative organisms including Pseudomonas aeruginosa or by staphylococci.

Infections of the eyes

Infections of the kidneys and/or the efferent urinary tract

Infections of the genital organs, including adnexitis, gonorrhoea, prostatitis
Infections of the abdominal cavity (e.g. infections of the gastrointestinal tract or of the biliary tract, peritonitis)

Infections of the skin and soft tissue

Infections of the bones and joints

Sepsis

Infections or imminent risk of infection (prophylaxis) in patients

whose immune system has been weakened (e.g. patients on immunosuppressants or have neutropenia)

Selective intestinal decontamination in immunosuppressed patients

For Ciprofloxacin film-coated tablets only:

Prophylaxis of invasive infections due to Neisseria meningitidis.

Children and adolescents

Ciprofloxacin may be used in children for the second- and third-line treatment of complicated urinary tract infections and pyelonephritis due to Escherichia coli (age range applied in clinical studies: 1 to 17 years) and for the treatment of bronchopulmonary infections of cystic fibrosis associated with Pseudomonas aeruginosa (age range applied in clinical studies: 5 to 17 years).

Treatment should only be initiated after careful benefit/risk evaluation, due to possible adverse events related to joints and/or surrounding tissues.

The clinical trials in children were performed in the indications listed above. For other indications clinical experience is limited.

Inhalational anthrax (post-exposure) in adults and in children (for Ciprofloxacin film-coated tablets only):

To reduce the incidence or progression of disease following exposure to aerosolized Bacillus anthracis.

CONTRAINDICATIONS:

Ciprofloxacin is contraindicated when there is a history of hypersensitivity to it or to other drugs of the quinolone class. It is also contraindicated in patients with a history of tendonitis or tendon rupture, in growing children and in pregnant or nursing mothers.

PRECAUTIONS /WARNINGS:

Like other fluoroquinolones, tendonitis incidents following ciprofloxacin treatment have been reported usually concerning the Achilles tendon and occasionally causing tendon rupture. If tendonitis develops, treatment should be discontinued and the patients immobilized. Factors contributing to tendonitis are age over 60 years, vigorous exercise and long-term corticosteroid treatment, as well as the initial recovery period of aged patients. Patients should be warned of the possibility of pain of the Achilles tendon. Ciprofloxacin administration has caused lameness in immature dogs. Histopathological examination of weight-bearing joints in these dogs showed permanent damage to the cartilage. Related drugs, such as nalidixic acid, cinoxacin and norfloxacin, also caused damage to weight-bearing joint cartilage and other arthropathic symptoms to immature animals of various species.

Like all other quinolones, ciprofloxacin can cause excitation of the CNS leading to tremor, restlessness, light-headedness, confusion and rarely illusions or convulsions.

Ciprofloxacin should therefore be used with caution in patients with known or suspected CNS disorders such as severe brain arteriosclerosis or epilepsy, or in patients with other factors that predispose to convulsions.

Ciprofloxacin crystals are rarely observed in human urine, but more often in laboratory animal urine, which is more alkaline than human urine. Patients should be well hydrated to avoid highly concentrated urine, and the alkalinity should be avoided. An adequate fluid intake should be maintained during ciprofloxacin to minimize the risk of crystalluria.

Aortic aneurysm and dissection

Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis).

In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department.

Dosage adjustment may be necessary in patients with renal insufficiency (see "Dosage"). As with other antibacterial agents, prolonged use of ciprofloxacin may cause overgrowth of nonsusceptible organisms. If superinfection occurs, discontinue the drug immediately.

Use in pregnant women: Reproduction toxicity studies in animals did not reveal any teratogenic effects, reduction of fertility or embryotoxicity. However, there are no adequate and well-controlled studies in pregnant women.

Ciprofloxacin should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

Use in nursing women: Ciprofloxacin is secreted in human and rat milk, like other drugs of this class. For this reason, and due to the possibility of serious adverse effects from ciprofloxacin to nursing infants, the discontinuation of either nursing or ciprofloxacin administration should be decided, taking into consideration the importance of the drug to the mother.

Use in children: Ciprofloxacin should not be used in children and adolescents in the growing phase, as safety and effectiveness in these patients have not been established. Fluoroquinolones have been shown to cause arthropathy in immature animals.

Effect of ciprofloxacin on ability to drive: Ciprofloxacin affects response speed to such a degree that the person's ability to drive and operate heavy machinery is affected. The effect is enhanced by alcohol.

Hypersensitivity & Skin Reaction:

Excessive sunlight may cause phototoxicity.

Interactions with other drugs or substances: Concurrent administration of ciprofloxacin with theophylline may lead to increased serum concentrations of theophylline and prolongation of its elimination of half-life. This may lead to increased risk of theophylline-related adverse reactions. If concurrent administration cannot be avoided, serum levels of theophylline should be monitored and dosage adjustments made.

Concurrent oral administration of ciprofloxacin with iron, antacids and antiretroviral drugs containing magnesium, aluminium or calcium reduces ciprofloxacin absorption, resulting in lower serum and urine levels. Ciprofloxacin should be administered 1-2 hours prior to or at least 4 hours after these compounds. This is not necessary in the case of H2 receptor inhibitors.

Probenecid interferes with renal tubular secretion of ciprofloxacin and produces increased serum ciprofloxacin levels. This should be considered if patients are receiving both drugs. As with other broad-spectrum antibiotics, prolonged use of ciprofloxacin can result in overgrowth of non-susceptible organisms. Repeated evaluation of patient condition and microbial susceptibility testing are essential. If superinfection occurs during therapy, appropriate measures should be taken.

Animal studies have shown that combining high doses of quinolones and certain nonsteroidal anti-inflammatory drugs (not acetylsalicylate acid) can lead to convulsions. Concurrent administration of ciprofloxacin with warfarin can enhance the latter's anticoagulant activity.

Transiently increased serum creatinine levels have been observed on concomitant administration of ciprofloxacin and cyclosporin. Therefore, serum creatinine levels must be tested regularly (twice a week). Concurrent administration of ciprofloxacin and cyclosporin may increase the risk of nephrotoxicity.

In some cases, concurrent administration of ciprofloxacin and glibenclamide can enhance glibenclamide activity (hypoglycaemia). Metoclopramide accelerates ciprofloxacin absorption, and peak plasma concentrations are achieved in less time. No effects on its bioavailability were observed. For systemically administered levofloxacin-containing products, the listing of risk factors should additionally include: "in patients receiving daily doses of 1000mg levofloxacin".

SPECIAL WARNING AND PRECAUTION FOR USE

The use of ciprofloxacin should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products (see section Undesirable Effects). Treatment of these patients with ciprofloxacin should only be initiated in the absence of alternative treatment options and after careful benefit/risk assessment.

Musculo-skeletal system

The risk of developing fluoroquinolone-associated tendonitis and tendon rupture is further increased in people older than 60, in those taking corticosteroid drugs, and in kidney, heart and lung transplant recipients. Patients experiencing pain, swelling, inflammation of a tendon or tendon rupture should be advised to stop taking their fluoroquinolone medication (Ciprofloxacin) and to contact their health care professional promptly about changing their antimicrobial therapy. Patients should also avoid exercise and using the affected area at the first sign of tendon pain, swelling or inflammation.

Exacerbation of myasthenia gravis

Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been associated with fluoroquinolones in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.

Prolonged, disabling and potentially irreversible serious adverse drug reactions

Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been



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Size : 140 x 440 mm
Export : Malaysia
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BS-01/08/24-V07
DG-09/11/22-V06
YP-26/11/20-V05
DG-21/11/20-V04
DG-12/11/20-V03
YP-14/11/19-V02
SS-07/11/19-V01



reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors. Ciprofloxacin should be discontinued immediately at the first sign or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.

Tendinitis and tendon rupture

Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometime bilateral, may occur as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age) with renal impairment, patients with solid organ transplants, and those treated concurrently with corticosteroids. Therefore, concomitant use of corticosteroids should be avoided. At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with ciprofloxacin should be discontinued and alternative treatment should be considered. The affected limb(s) should be appropriately treated (e.g. immobilisation). Corticosteroids should not be used if signs of tendinopathy occur.

Peripheral neuropathy

Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesia, hypaesthesia, dysesthesia, or weakness have been reported in patients receiving quinolones and fluoroquinolones. Patients under treatment with ciprofloxacin should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of potentially irreversible condition (see section Undesirable Effects).

Psychiatric reactions

Psychiatric reactions may occur even after the first administration of fluoroquinolones, including Ciprofloxacin. In rare cases, depression or psychotic reactions can progress to suicidal ideations/thoughts and self-injurious behaviour, such as attempted or completed suicide (see section 'Undesirable effects'). In the event that the patient develops these reactions, Ciprofloxacin should be discontinued and appropriate measures instituted. Caution is recommended if Ciprofloxacin is to be used in psychotic patients or in patients with a history of psychiatric disease.

Dosage regimen

Unless otherwise prescribed, the following daily doses are recommended for Adults

Table 1: Recommended daily doses of Ciprofloxacin oral in adults

Indications	Daily dose of ciprofloxacin in mg for Ciprofloxacin film-coated tablets
Infections of the respiratory tract I (according to severity and organism)	2 x 500 mg to 2 x 750mg
Urinary tract infections: - acute, uncomplicated - cystitis in women (before menopause) - complicated	2 x 250 mg to 2 x 500 mg single dose 500 mg 2 x 500 mg to 2 x 750 mg
Genital infections - uncomplicated gonorrhea (including extragenital sites of infection) - adnexitis, prostatitis, epididymo-orchitis	1 x 500 mg 2 x 500 mg to 2 x 750 mg
Diarrhea	2 x 500 mg
Other infections (see indications)	2 x 500 mg
Particularly severe, life-threatening infections, i.e. - Recurrent infections in cystic fibrosis - Bone and joint infections - Septicemia - Peritonitis. In particular when <i>Pseudomonas</i> , <i>Staphylococcus</i> or <i>Streptococcus</i> is present	2 x 750 mg
Inhalational anthrax (post exposure)	2 x 500 mg
Prophylaxis of invasive infections due to <i>Neisseria meningitidis</i>	1 x 500 mg as a single dose

Children and adolescents

Table 2: Recommended daily doses of Ciprofloxacin oral in children and adolescents

Indications	Daily dose of ciprofloxacin in mg for Ciprofloxacin oral
Infections in cystic fibrosis	2 — 20mg/kg body weight (maximum of 750 mg per dose)
Complicated urinary tract infections and pyelonephritis	2 — 10mg/kg body weight to 2 — 20mg/kg body weight (maximum of 750 mg per dose)
Inhalational Anthrax (Post-exposure)	2 — 15mg/kg body weight (maximum of 500mg per dose)

Missed dose

If a dose is missed, it should be taken as soon as the patient remembers and then treatment should be continued as prescribed. Double doses should not be taken to compensate for a missed dose.

Additional information on special patient population

Children and adolescents

For recommended dosage regimen see Table 2.

Geriatric patients (> 65 years)

Elderly patients should receive a dose as low as possible depending on the severity of their illness and the creatinine clearance. (see also *Patients with renal and hepatic impairment*)

Patients with renal and hepatic impairment

Adults

Patients with renal impairment

Table 3: Recommended doses for patients with renal impairment

Creatinine Clearance [mL/min/1.73 m ²]	Serum Creatinine [mg/100mL]	Total daily oral dose of ciprofloxacin
30 to 60	1.4 to 1.9	maximum 1000 mg
below 30	2.0	maximum 500mg

- Patients with renal impairment on hemodialysis
 - For patients with creatinine clearance between 30 and 60 mL/min/1.73m² (moderate renal impairment) or serum creatinine concentration between 1.4 and 1.9 mg/100 mL, the maximum daily oral dose of ciprofloxacin should be 1000 mg or 800 mg for an intravenous regimen.
 - For patients with creatinine clearance less than 30 mL/min/1.73m² (severe renal impairment) or serum creatinine concentration equal or higher than 2.0 mg/100 mL, the maximum daily oral dose of ciprofloxacin should be 500 mg or 400 mg for an intravenous regimen on dialysis days after dialysis
- Patients with renal impairment on continuous ambulatory peritoneal dialysis (CAPD)
 - Addition of Ciprofloxacin solution for infusion to the dialysate (intraperitoneal): 50 mg ciprofloxacin / liter dialysate administered 4 times a day every 6 hours
 - The maximum daily oral dose of ciprofloxacin should be (1 x 500 mg Ciprofloxacin film-coated tablet or 2 x 250 mg Ciprofloxacin film-coated tablets).
- Patients with hepatic impairment
 - In patients with impaired hepatic function no dose adjustment is required
- Patients with renal and hepatic impairment
 - For patients with creatinine clearance between 30 and 60 mL/min/1.73m² (moderate renal impairment) or serum creatinine concentration between 1.4 and 1.9 mg/100 mL, the maximum daily oral dose of ciprofloxacin should be 1000 mg or 800 mg for an intravenous regimen.
 - For patients with creatinine clearance less than 30 mL/min/1.73m² (severe renal impairment) or serum creatinine concentration equal or higher than 2.0 mg/100 mL, the maximum daily oral dose of ciprofloxacin should be 500 mg or 400 mg for an intravenous regimen.

Children and adolescents

Dosing in children with impaired renal and or hepatic function has not been studied.

Inhalational Anthrax (Post-exposure) in Adults and Children

For recommended dosage regimen see Table 1 (for adults) and Table 2 (for children and adolescents), respectively.

Dosage adjustments:

Dosages usually do not need to be adjusted, except under the

following conditions:

- Reduced renal function: Creatinine clearance <20ml/min or serum creatinine > 3mg/100ml. Dose should be reduced to 50% of the total daily dose.
- Reduced renal function and haemodialysis: Dose as above. Dose should be administered after dialysis.
- Reduced renal function + CAPD (continuous peritoneal dialysis in outpatients): Administration of 500mg ciprofloxacin orally 4 times a day (every 6 hourly)
- Reduced hepatic function: No dosage adjustment required.
- Reduced renal and hepatic function: Dosage adjustment as for reduced renal function.

Elderly patients should be given as low a dose as is necessary, depending on the severity of their illness and their renal function status.

ADVERSE REACTIONS:

Hepatic insufficiency, renal insufficiency, anaphylactic reactions, hemolysis and hypoglycaemia (especially when concomitantly administered with glibenclamide) can occur upon quinolone therapy and could be life-threatening, if not treated.

Also tendonitis, especially of the Achilles tendon, that can lead to tendon rupture.

Patients concurrently receiving coumarin anticoagulants must be carefully monitored because of the risk of increased anticoagulant activity.

Ciprofloxacin is generally well-tolerated. Most common effects, either related to the drug or not, were nausea, diarrhea, vomiting, abdominal pain or discomfort, headache, restlessness and rash. Adverse effects observed in rare cases are reported below.

Gastrointestinal: Adverse gastrointestinal symptoms occur most commonly (up to 10% of patients treated), and usually comprise of nausea, diarrhea, vomiting, dyspepsia, abdominal pain, flatulence, anorexia, which are generally mild and transient.

In case of severe diarrhea during or after treatment, the patient must be examined for pseudomembranous colitis. In this case, ciprofloxacin must be discontinued and suitable treatment established (e.g. vancomycin 4 x 250 mg a day, orally). Drugs that interfere with peristalsis should be avoided.

Oral ulceration, oral candidiasis, dysphagia, intestinal perforation and gastrointestinal bleeding have occurred rarely.

Central Nervous System: Stimulatory effects on the CNS are seen in 1-4% of patients, but are usually mild. The effects observed include vertigo, headaches, weakness, restlessness, tremor. Rarely, light-headedness, perspiration, balance disorders, insomnia, nightmares, illusions, manic reactions, irritability, ataxia, convulsions, lethargy, somnolence, weakness, malaise, anorexia, phobia, depersonalization, depression, hallucinations, increased intracranial pressure, stress, confusion, painful peripheral paraesthesia.

In certain cases, these reactions are seen after administration of the first dose. In these cases, ciprofloxacin should be discontinued and the patient should remain under observation.

Hypersensitivity reactions, most commonly skin reactions such as rashes, urticaria, pruritus, have been reported in about 1% of patients and are usually mild to moderate in severity.

Rarely: petechiae, haemorrhagic blisters and vesicles, skin coloring, erythema nodosum, erythema multiforme, skin candidiasis, flushing, photosensitivity, Steven-Johnson syndrome, Lyell syndrome. Analaphylactic reactions are occasionally seen after the first administration of ciprofloxacin. Interstitial nephritis, hepatitis, hepatic necrosis that very rarely evolves into life threatening hepatic insufficiency. When these conditions occur, ciprofloxacin should be discontinued and suitable therapy established.

Special senses: Rarely, blurred vision, visual disorders (color perception anomalies, overbrightness of lights), reduced visual acuity, diplopia, eye pain, tinnitus, transient hearing loss (especially high frequency), bad taste, anosmia.

Musculoskeletal: Rare arthralgias or back pain, joint stiffness, joint edema, neck or chest pains, flare up of gout, muscle pains. In isolated cases, tendonitis of the Achilles tendon occurred. Also, partial or complete rupture of the tendon occurred, particularly in elderly patients who earlier received glucocorticoids regularly. For this reason, if tendonitis is suspected (painful edema), ciprofloxacin should be discontinued.

Renal-Urogenital: Rarely, interstitial nephritis, nephritis, renal insufficiency, polyuria, urine retention, vaginitis, urethral bleeding, acidosis.

Cardiovascular: Palpitation and rare instances of auricular flutter, ventricular ectopia, syncope, hypertension, angina pectoris, myocardial infarction, cardiopulmonary arrest, cerebral thrombosis, migraine, collapse.

Respiratory: Most of these incidents were mild or moderate, regressed soon after the drug was discontinued and required no therapy. In some cases nausea, vomiting, tremor, restlessness and palpitation were considered by the researchers to be due to high plasma levels of theophylline, possibly due to interactions between the drug and ciprofloxacin.

Adverse laboratory changes: Laboratory parameter changes, recorded as adverse effects, regardless of drug relationship:

Hepatic: increase in SGPT (ALT), SGOT(AST), alkaline phosphatase, LDH, serum bilirubin.

Haematologic: Eosinophilia, leucopenia, agranulocytemia, lower platelet count, higher red blood cell count, pancytopenia.

Renal: Serum creatinine increase, blood urea nitrogen increase. Crystalluria, cylinduria and haematuria have been reported.

Other changes, reported in rare instances, were: increase in gamma-GT, increased serum amylase, decrease in blood sugar, increase in uric acid, lower hemoglobin levels, anaemia, increased number of large mononucleates, leucocytosis, bleeding diathesis, haemolytic anaemia, changes in prothrombin time.

Musculoskeletal and connective tissue disorder*
Nervous system disorders*
General disorders and administrative site conditions*
Psychiatric disorders*
Eye disorders*
Ear and labyrinth disorders*

*Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendinitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in patients receiving fluoroquinolones irrespective of pre-existing risk factors (see section Warnings and Precautions).

Psychiatric disorders

Rare: Depression (potentially culminating in suicidal ideations/ thoughts or suicide attempts and completed suicide)

Very Rare: Psychotic reactions (potentially culminating in suicidal ideations/ thoughts or suicide attempts and completed suicide)

Post Marketing Experience: Exacerbation of myasthenia gravis

OVERDOSAGE AND TREATMENT:

In cases of overdosage, reversible nephrotoxicity has been occasionally reported.

Therefore, besides normal emergency measures, it is suggested that renal function, including urinary pH and acidity, be monitored and antacids containing magnesium or calcium, which reduce ciprofloxacin absorption, be administered. Only a small fraction (<10%) of ciprofloxacin is removed from the body by hemodialysis or peritoneal dialysis.

In the case of renal damage resulting in prolonged oliguria, dialysis should be initiated.

Serum level of ciprofloxacin are reduced by dialysis.

Adequate hydration should be maintained to minimize the risk of crystalluria.

STORAGE:

Store below 30°C. Protect from light.

Shelf-life: 3 years

Keep all medicines out of reach of children.

Date of revision: 01 Aug 2024

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Manufactured by:

 **CADILA**
PHARMACEUTICALS
L I M I T E D

1389, Trasad Road, Dholka-382225,
Dist: Ahmedabad, Gujarat State, India.



Back Side

Size : 140 x 440 mm
Export : Malaysia
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DG-21/11/20-V04
DG-12/11/20-V03
YP-14/11/19-V02
SS-07/11/19-V01